

Description of a Linear Programming Approach for a Water Right Apportionment Solver

Background

After-the-fact distribution accounting — the process of subdividing and naming measured diversions into components representing water diverted under direct flow rights, storage deliveries, and other such *transactions* — is key to documenting and reporting water distribution in Utah river systems. The transparency of this accounting will become increasingly important as public scrutiny of water management increases.

Distribution accounting is not a new concept. The Utah Division of Water Rights (DWRi) and other state engineer offices in western states have performed this type of accounting for decades. In Utah, this accounting is done using system-specific calculation routines. Although the principles of priority distribution are constant from basin to basin, each river system has a unique hydrography, flow measurements, water rights, and other water transactions; as a result, the current approach to distribution accounting involves developing a unique and customizable calculation routine for each river system.

There are major problems with the current approach:

1. It takes significant time and resources to develop system-specific accounting calculation routines.
2. Much of the information in the calculation routines is duplicated. Embedded in these formulas is information about water rights (their locations, priorities, and flow limitations) and the stream hydrology (locations of stream gages and reaches). Because of this duplication, inconsistencies arise with how a water right is represented in distribution accounting versus the official water right records.
3. Neither the data used for accounting calculations or the accounting results are readily understandable or reusable. Although the underlying information embedded in calculation routines is of general interest and could be useful for many purposes, this information is inaccessible and opaque when embedded in calculation formulas. Because it's not standardized, the accounting results themselves may mean somewhat different things from system to system. There is a need for all of this information to follow standards and be accessible and understandable.
4. Staff responsible for developing and maintaining accounting calculation routines must have a very unique set of expertise and experience to interpret water rights, understand local distribution practices, and write computer logic. This dependence on highly skilled staff may be unsustainable.

These problems can be solved. The proposed approach organizes the task into several independent work efforts and datasets. These datasets can be managed separately by the teams best positioned to maintain them — there is no need for a single employee to do all of this work. The work is organized to avoid duplication and the resulting potential for inconsistencies. These datasets can be reused for a variety of

purposes, both within the division and by outside users. This will result in greatly improved data management. The components of the proposed workflow include:

“The Water Rights Network Hydrography” is a dataset that maps the location and connectivity of streams and diversions like pipelines, canals, and ditches. These features can be used as a referencing system to precisely define the location of measurements and the paths of water rights and other transactions.

“The General Transaction Schedule” is a database of authorized water diversions and deliveries and other transactions. Many distribution systems have priority schedules or delivery schedules that contain similar information. The general transaction schedule will standardize these datasets. The path of each transaction, where water is moved from beginning point to end point, will be represented on the water rights network to make the distribution aspects of water rights and other transactions machine readable.

“The List of Accounts” is part of an accounting database that defines accounts that transactions move water between.

“The Observation Database” is just what it sounds like — a database housing measurement data for diversions, stream flow, evaporation, and other data needed for accounting and other purposes. Utah State University is currently preparing a report that will recommend improvements to DWRI’s existing DVRT database.

“The General Apportionment Solver” is the subject of this document. It is not a dataset, but rather a program that calculates apportionments using these other dataset components. All of the information the solver needs is contained in these datasets, so the apportionment calculations are defined without staff needing to write and maintain logic for a system-specific accounting model. In other words, the general solver eliminates the duplication of this information within system-specific accounting model code.

Formulation

The general apportionment solver works by converting the input data into a system of linear equations that can be solved to obtain the apportionment values. This involves:

1. Simplifying the hydrography into an accounting flow network,
2. Defining variables to represent transaction apportionments,
3. Defining constraints, and
4. Formulating a sequence of linear optimization objectives.

1) Accounting Flow Network

The stream and canal hydrography lines provide flow network information that is required for the apportionment calculation problem. But we do not have enough measurements or knowledge regarding stream gains and losses to manage the system at the level of every individual flowline segment as defined in the hydrography dataset. It is not always practical or necessary to account for water at the finest level of detail. For accounting purposes, we simplify and abstract the stream and canal hydrography into an

accounting flow network. This accounting flow network is similar to the base hydrography in that it is composed of directed flowlines and nodes, but these accounting flow network elements do not necessarily represent discrete streams or canals — they may be more abstract.

Figure 1 shows how the flowlines in the system hydrography (left) are simplified into an accounting flow network (right).

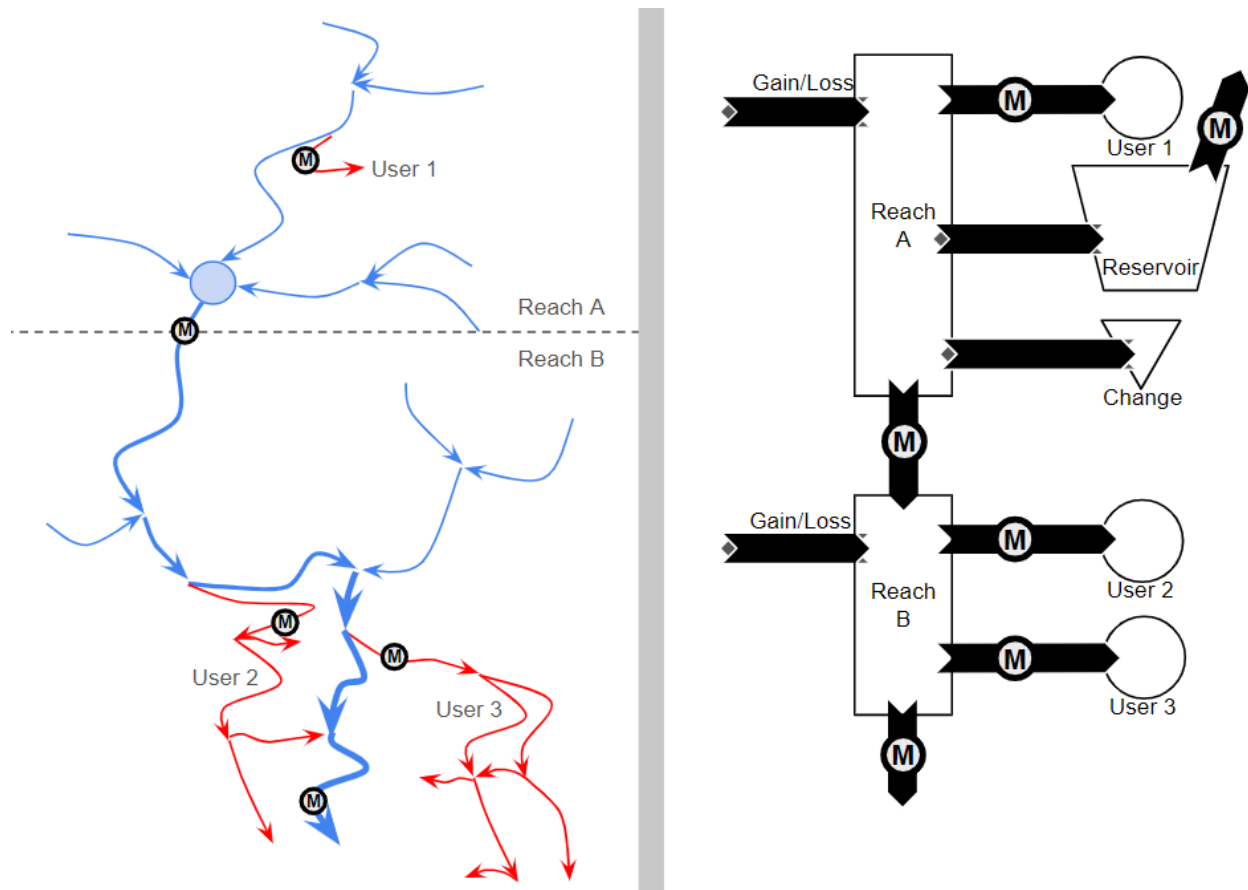


Figure 1. Visualization of how the Water Right Network hydrography (left) is simplified into the accounting flow network (right) for use by the general apportionment solver.

The accounting flow network may include the following elements:

“*Reach nodes*” represent a segment of a natural stream. Two reaches on the same river system are separated by a stream gage. A river system can have any number of reaches, but locations where the stream is bisected into separate reaches must always correspond to a measurement. Each reach has a gain/loss flowline, with positive flows representing gains and negative flows representing losses.

“*Use nodes*” represent the destination for most diversions and deliveries. For general distribution accounting, we are not interested in the complexities of a canal company’s distribution system, including whether water is conveyed from one canal company to another, the location and timing of returns, etc.

“*Storage nodes*” represent water diverted into a storage reservoir where it may be released back into a natural channel to make deliveries. Although on-stream reservoirs are physically collated with the natural stream and within a stream reach, representing them as a distinct storage node is an abstraction that allows the diversion into storage to be represented. An off-stream reservoir that releases water into a natural channel to make deliveries should also be represented as a storage node. For on-stream reservoirs, each storage node has a flowline originating from a reach node; the convention is that positive flows represent net flow from the reach into the storage node and negative flows represent deliveries from to the reach that exceed diversions from the reach into the node.

“*Change nodes*” are an abstract construct that assist in the accounting of change transactions. A base water right transaction that has been subsequently changed would flow into a change node with the priority of the base water right. A change transaction with the priority of the change would then flow out from this change node. The change node serves as a handoff cache between these related transactions, somewhat similar to how a storage node is an intermediary between transactions representing diversions into storage and storage deliveries. Each change node has a flowline originating from a reach node that represents both inflow into and outflow from the change node.

“*Flowlines*” in the accounting flow network represent the potential for water to be moved from one node to another. Only flowlines having actual flows that are measured, estimated, or can be calculated with a mass balance equation are included. Multiple flowlines can exist between the same nodes, for example if a user has multiple diversions that are metered individually. Generally flowlines must have non negative flows, but certain types of flowlines are allowed to have negative or backward flows. (For example, a flowline representing the loss from a reach will be negative, and flowlines representing net releases from an on-stream storage node back to the reach node will be negative.) The total actual flow along a flowline is known; what needs to be solved are the individual apportionments that sum together to comprise the total flow along some of these flowlines.

2) Variables

Each flowline of the simplified flow network graph is a possible route for transactions. Since we are interested in quantifying the transactions (most of the flows between nodes are measured), our variables represent flow for each transaction, not the total flow along a flowline. There may be many transaction variables passing through a flowline. This is shown by Figure 2.

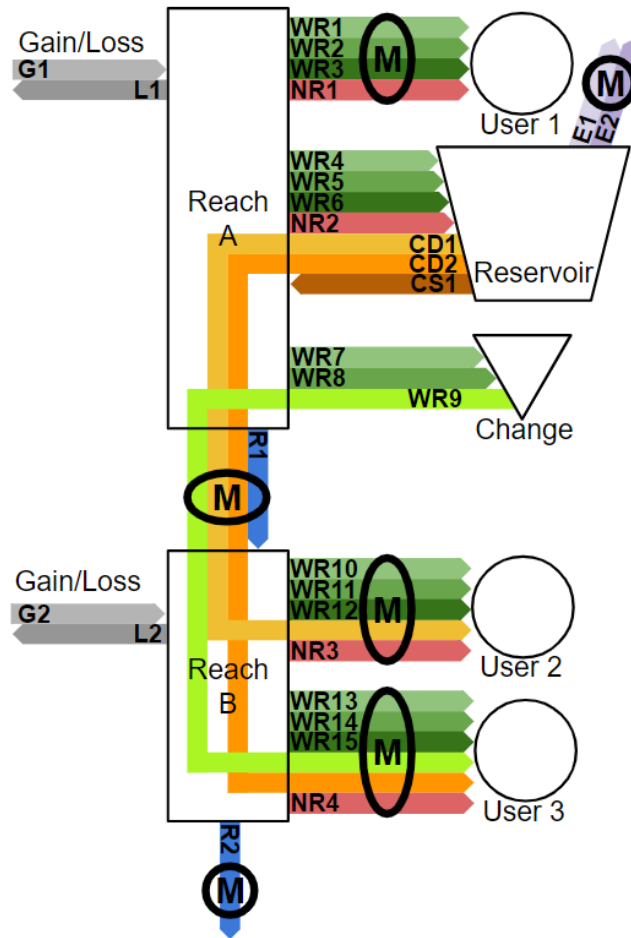


Figure 2. Example schematic showing how transaction variables could follow routes along the accounting flow network shown in Figure 1. The different arrows of various colors represent different transactions. The objective of the general apportionment solver is to calculate the flow for each transaction.

As illustrated by this figure, there are several classes of transaction variables:

“Reach gain/loss” variables (G/L prefix) are the only variables that pass over a gain/loss flowline. They move water between a single reach and outside the system. They are positive unbounded variables.

“Reach-to-reach” variables (R prefix) represent the remaining portion of flow from an upstream reach to a downstream reach that is not apportioned to a prioritized transaction (storage delivery, change/exchange delivery, etc.). If a large, senior right is located at the bottom of a river system, these reach-to-reach flows will be large, signifying that a large portion of the natural flow available in upstream reaches is being delivered to the downstream reaches. They are positive unbounded variables.

“Water right diversion” variables (WR prefix) move water from a reach node or change node into a use or storage or change node. These transactions are explicitly defined in the transaction schedule along with a maximum cfs limit and a priority order. There are typically many water rights authorizing diversions into

any particular canal, and each water right gets its own variable. They are positive variables that generally have an upper-bound corresponding to the water right limit.

“*Unauthorized diversion*” variables (NR prefix) represent water diverted from a reach without a water right. Water users and commissioners should avoid unauthorized diversion, but this class of transaction variables must be included to ensure the problem remains feasible when these unauthorized diversions do occur. Each diversion flowline from a reach node to a use or storage node must have an unauthorized diversion variable to ensure the problem remains feasible. They are positive unbounded variables.

“*Controlled delivery*” variables (CD prefix) begin at a storage node or as an import from another system and may traverse through any number of reaches, before ending at a use node or another storage node. These transactions are also explicitly defined in the transaction schedule, with a priority order that is later than all water right diversion transactions. As on the water rights network, these delivery paths are not limited to traveling in the same direction as the natural stream -- they may go in the opposite direction as well, so long as the path from beginning to end is continuous. And deliveries from a reservoir that are not distributed through a natural channel do not need to be included as variables since we are generally concerned only with apportioning the water in natural streams. They are positive unbounded variables.

“*Controlled spill*” variables (CS prefix) represent the typically unintentional loss of controlled storage or import water into a reach. These paths begin at a storage node or as an import from another system and end at a reach node. For example, if an on-stream reservoir releases storage but no users with storage contracts take the delivery, the water must be accounted for as a controlled spill variable. After being lost from the control of the previous water user, this water is then apportioned to existing rights as if it were natural flow. Water users generally manage their systems to avoid this type of loss, but this class of variables must be considered to ensure the problem remains feasible when these abnormal conditions occur. Each flowline that is used to convey controlled water to a reach node must have an unauthorized diversion variable to ensure the problem remains feasible. They are positive unbounded variables.

“*Storage evaporation*” variables (E prefix) represent evaporative losses from storage nodes. Evaporative losses may be apportioned to many distinct storage accounts in the reservoir, and each apportionment is represented with a separate storage evaporation variable. These transactions are explicitly defined in the transaction schedule so the sequence and proportions can be appropriately set. They are positive unbounded variables.

3) Constraints

In addition to being constrained to being non-negative and possibly within an upper-limit, the transaction variables are also constrained by flow measurement equations and mass balance equations at each reach node, storage node, and change node. The accounting flow network makes it straightforward to identify the variables involved in these equations.

“*Flow measurement equations*” constrain each transaction variable passing the measurement. The measurement equals the sum of the transactions; if any transactions traverse in the opposite direction, those backward-directed variables are subtracted rather than added. The following equations represent the equations for the example shown in Figure 2:

$$\begin{aligned}
M_{User\ 1} &= x_{WR1} + x_{WR2} + x_{WR3} + x_{NR1} \\
M_{User\ 2} &= x_{WR10} + x_{WR11} + x_{WR12} + x_{SD1} + x_{NR3} \\
M_{User\ 3} &= x_{WR13} + x_{WR14} + x_{WR15} + x_{WR9} + x_{SD2} + x_{NR4} \\
M_{Reach\ A\ to\ B} &= x_{WR9} + x_{SD1} + x_{SD2} + x_{R1} \\
M_{Reach\ B\ out} &= x_{R2}
\end{aligned}$$

“*Reach mass balance equations*” constrain transactions involving diversions from natural reaches. Reach storage is assumed to be negligible, so inflow into the reach equals outflow from the reach. Transactions that pass through a reach have an inflow term and an outflow term, and thus cancel out.

$$\begin{aligned}
0 &= x_{G1} - x_{L1} - x_{R1} + x_{SS1} \\
&\quad + x_{WR1} + x_{WR2} + x_{WR3} + x_{NR1} \\
&\quad + x_{WR4} + x_{WR5} + x_{WR6} + x_{NR2} \\
&\quad + x_{WR7} + x_{WR8} \\
0 &= x_{G2} - x_{L2} + x_{R1} - x_{R2} \\
&\quad + x_{WR10} + x_{WR11} + x_{WR12} + x_{NR3} \\
&\quad + x_{WR13} + x_{WR14} + x_{WR15} + x_{NR4}
\end{aligned}$$

“*Storage mass balance equations*” are similar to the reach mass balance equations, but with measured change in storage.

$$\begin{aligned}
M_{\Delta S} &= x_{WR4} + x_{WR5} + x_{WR6} + x_{NR2} \\
&\quad - x_{SD1} - x_{SD2} - x_{SS1} - x_{E1} - x_{E2}
\end{aligned}$$

“*Change node mass balance equations*” are another type of constraint. These are adjusted during the process of calculating the apportionments, as discussed in the Objective Sequence For Water Right Changes section.

$$0 = x_{WR7} + x_{WR8} - x_{WR9}$$

4) Objective Sequence

The system of equations incorporating the constraint equations is generally ill-posed. There may be an infinite number of ways that values can be apportioned to the transactions. After all, there may be any number of water rights for any particular diversion, resulting in many more variables than constraint equations. The particular solution we seek is the one that follows the principle of prior appropriation — a more senior priority transaction should receive a full apportionment before a more junior transaction receives any.

There are a variety of techniques that could be applied to incorporate this priority information into the objective function. For example, weights could be used in the objective function to give preference to solutions that maximize senior variables. The senior-most variable would be given the largest weight, the next senior-most variable would be given the next largest weight, and so on. The approach described here, however, uses a sequence of objectives; the first objective would be to maximize the senior-most variable; the next objective would be to maximize the next senior-most variable, and so on. This approach not only allows calculation of unique-priority transactions but also provides flexibility that is needed to deal with equal-priority transactions and change transactions.

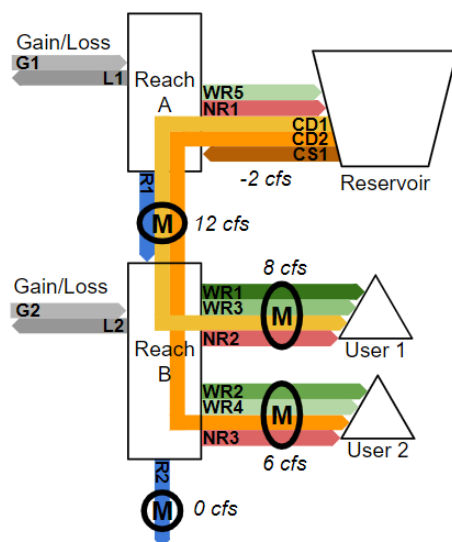
For Unique-Priority Transactions

For a collection of transactions each having a unique priority, the solution can be obtained by solving for each transaction in priority order:

1. First, the system of equations is solved with the objective of maximizing the most senior transaction.
2. That transaction is then converted from a variable to a constant known value, equal to the maximized value.
3. This process is then repeated for every transaction variable, in priority order, until they all have been quantified and eliminated as system unknowns.

An Example

In this example, there are 7 prioritized transactions requiring 7 steps to calculate an apportionment for each. There are two users, each with 2 water rights apiece and each entitled to storage deliveries. The water right for diverting water into the reservoir is the 7th variable. In this example, the reservoir outflow exceeds reservoir inflow by 2 cfs as determined by observing the storage change.



Variable	WR1	WR2	WR3	WR4	WR5	CD1	CD2
Limit	3	5	2	4	20	N.A.	N.A.
Priority	1st	2nd	3rd	4th	5th	6th	7th
Step #1	3						
Step #2	3	5					
Step #3	3	5	2				
Step #4	3	5	2	1			
Step #5	3	5	2	1	1		
Step #6	3	5	2	1	1	3	
Step #7	3	5	2	1	1	3	0

- In step #1, the most senior transaction (WR1) is maximized. The water right limit for WR1 is 3 cfs. The flow measurement to User 1 requires that WR1 cannot not exceed 8 cfs. The reach mass balance equations require that WR1 cannot exceed the natural flow available at Reach B, which includes 10 cfs of gains in Reach A and 2 cfs of gains in Reach B for a total of 12 cfs. So the governing constraint is the water right limit, and WR1 is set equal to 3 cfs.
- In step #2, the next senior-most transaction (WR2) is maximized. The water right limit for WR2 is 5 cfs. The flow measurement to User 2 prevents WR2 from exceeding 6 cfs, and there is 9 cfs of natural flow remaining ($12 \text{ cfs} - \text{WR1} = 9 \text{ cfs}$). So WR2 is set equal to 5 cfs.
- In step #3, WR3 is maximized subject to a water right limit (2 cfs), the remaining flow measurement to User 1 ($8 \text{ cfs} - \text{WR1} = 5 \text{ cfs}$), and the remaining natural flow ($12 \text{ cfs} - \text{WR1} - \text{WR2} = 4 \text{ cfs}$). The water right limit governs, and WR3 is set equal to 2 cfs.
- In step #4, WR4 is maximized subject to a water right limit (4 cfs), the remaining flow measurement to User 2 ($6 \text{ cfs} - \text{WR2} = 1 \text{ cfs}$), and the remaining natural flow ($12 \text{ cfs} - \text{WR1} - \text{WR2} - \text{WR3} = 2 \text{ cfs}$). The remaining flow measurement constraint governs, and WR3 is set equal to 1 cfs.
- In step #5, WR5 is maximized subject to a water right limit (20 cfs), the storage change measurement constraint, and the remaining natural flow in Reach A. These constraints are not as trivial to analyze because they are intertwined with the storage delivery variables and constraints. In any case, the solution is that WR5 is maximized and set equal to 1 cfs without violating the system of equations.
- In step #6, SD1 representing the storage delivery to User 1 is maximized subject to the reservoir storage change constraint (reservoir deliveries must equal 3 cfs, so SD1 cannot exceed 3 cfs) and the User 1 diversion measurement constraint ($8 \text{ cfs} - \text{WR1} - \text{WR3} = 3 \text{ cfs}$). So SD1 is set equal to 3 cfs.
- In step #7, SD2 representing the storage delivery to User 2 is maximized subject to the reservoir storage change constraint (reservoir deliveries must equal 3 cfs, so SD2 cannot exceed 3 cfs - $\text{SD1} = 0 \text{ cfs}$) and the User 2 diversion measurement constraint ($6 \text{ cfs} - \text{WR2} - \text{WR4} = 0 \text{ cfs}$). So SD1 is set equal to 0 cfs.

Now that these 7 variables have been solved for, we are left with 10 remaining non-prioritized variables, specifically: NR1, NR2, NR3, CS1, G1, L1, G2, L2, R1, R2. But some of these variables can be combined. What we are really interested in is the net gains in each reach, so G1-L1 and G2-L2. Similarly, we just need to know NR3-CS1 and can define these transactions such that only one may be non-zero. With these simplifications, we now have 7 unknowns and 7 equations and can solve for the remaining unknowns.

For Equal-Priority Transactions

If multiple transactions are assigned the same priority, we are looking for the unique solution where, to the extent possible, the delivered water is apportioned between each of these transactions in proportion to

their cfs limit. The procedure for obtaining this solution involves variable substitution and maximization calculations:

1. First, each of the same-priority variables are replaced by a factored replacement variable. The replacement variable is factored in a way that results in apportionments that are in proportion to the water right limits.
2. The system is then solved with the objective of maximizing this new replacement variable.
3. The original transaction variables are then substituted back into each equation, and each is assigned a new lower-bound.
4. Each same-priority variable is inspected to see if it can be increased further. Variables that are completely maximized and cannot receive any additional apportionment are fixed.
5. The process is then repeated, without the variables that cannot be increased any more, until all equal-priority variables have been maximized.

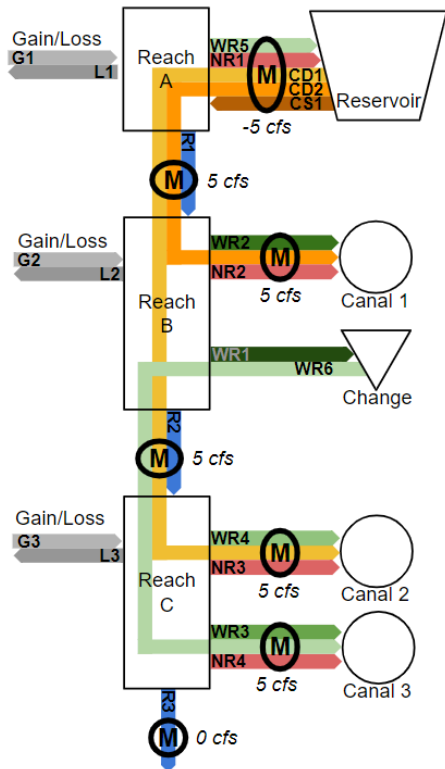
For Water Right Changes

The procedure must ensure changes do not impact how apportionments are calculated for transactions that are senior to the change. This is where change nodes come into play.

1. The change node constraint starts off relaxed, meaning inflow may exceed outflow into the change node.
2. The sequence of apportionment calculations proceeds as described previously for ordered priority and/or equal priority transactions, including the calculation of transaction variables that represent change base rights which flow into change nodes.
3. The sequence is interrupted immediately after the most junior transaction leaving the change node receives its apportionment. At this point, the change node constraint is adjusted to require inflow to equal outflow at the change node.
4. If this constraint tightening requires reducing the inflow into the change node, meaning the apportionment to the change base rights must be reduced, there is now additional water that should have been available to other transactions. We must loop back and recompute the apportionments for these other transactions with this updated information.

Change Example

In this example there are 3 reaches, one storage reservoir, 3 canal diversions, and one change. Each canal diverted 5 cfs, and reservoir measurements suggest the reservoir released 5 cfs. The measurements suggest Reach A has no gain or loss, Reach B gains 5 cfs, and Reach C gains 5 cfs. In this example, each of the water rights has a limit that is greater than 5 cfs, so the water right limits don't come into play.



Variable	WR1	WR2	WR3	WR4	WR5	WR6	CD1	CD2
Priority	1st	2nd	3rd	4th	5th	6th	7th	8th
Step #1	5							
Step #2	5	0						
Step #3	5	0	5					
Step #4	5	0	5	0				
Step #5	5	0	5	0	0			
Step #6	5	0	5	0	0	0		
Step #7	0					0		
Step #8	0	5				0		
Step #9	0	5	5			0		
Step #10	0	5	5	0		0		
Step #11	0	5	5	0	0	0		
Step #12	0	5	5	0	0	0	5	
Step #13	0	5	5	0	0	0	5	0

Prior to step #1 the change node constraint is relaxed so WR1 does not have to equal WR6. Steps #1-6 apportion water to the transactions under this relaxed condition:

- In step #1, WR1 receives 5 cfs because that is what is available as natural flow at Reach B.
- In step #2, WR2 receives 0 cfs because there is no remaining natural flow at Reach B after WR1 takes the 5 cfs.
- In step #3, WR3 receives 5 cfs because there is 5 cfs available as natural flow at Reach C and that is what was measured to Canal 3. (We don't bother to note that WR6 can deliver 5 cfs that was deposited by WR1 because we are imagining that the change that created WR6 has not yet come into existence, just like we are ignoring the other junior transactions during each step.)
- In step #4, WR4 receives 0 cfs because there is no remaining natural flow at Reach C. WR3 took it all.
- In step #5, WR5 receives 0 cfs because there is no natural flow at Reach A.
- In step #6, WR6 receives 0 cfs because it is not needed. Canal 3 only took 5 cfs, which the apportionment to WR3 covers.

After WR6 receives its apportionment, the change node constraint is then applied, meaning WR1 must equal 0 cfs and there is now 5 cfs of additional water that could have been apportioned to other transactions. So steps #7-11 repeat the calculation of apportionments for these variables.

- In step #7, WR1 is reevaluated and now cannot exceed WR6, so it must be 0 cfs.
- In step #8, WR2 is reevaluated and now receives the 5 cfs of available natural flow at Reach B.

- In steps #9-11, WR3, WR4, and WR5 receive the same apportionment as before and for the same reasons.

Finally, apportionments are assigned to CD1 and CD2. And then the remaining non-prioritized variables can be solved for.

- In step #12, CD1 receives an apportionment of 5 cfs, limited by the measured diversion at Canal 2.
- In step #13, CD2 receives an apportionment of 0 cfs because the diversion at Canal 1 is fully apportioned to WR2 and because the release from the reservoir is fully apportioned to CD1.

This example is a case where having a change node makes a difference in the accounting. If we modify the example so transaction WR1 goes from Reach B to Canal 3, WR1 would receive an apportionment of 5 cfs, WR2 would receive an apportionment of 0 cfs, and the 5 cfs storage delivery would be apportioned to Canal 2 (CD2) instead of Canal 1 (CD1).

Other Considerations

Storage Evaporation Apportionments

- Often the evaporation is divided proportionally between the storage accounts
- Should the pre- or post-delivery account balances be used? Can the general solver accommodate either method?
- Negative storage account balances - should these accounts get an evaporation credit?
- Do these questions belong to the domain of the general apportionment solver?

Storage Delivery Losses

In Utah, some storage deliveries are charged a percentage loss. This loss reflects the portion of the water released for a particular delivery that cannot be re-diverted into the user's canal. The loss is defined as a percentage of the storage release.

- what transaction variables would be needed
- how they would be solved with the general solver

Some states employ more detailed approaches to estimate and account for transit losses of water transactions that utilize the natural stream channel. These losses may include multiple components such as channel storage, bank storage, inadvertent diversions, and evaporation. These losses may be estimated using modeling studies. Although these losses depend on a wide variety of highly variable conditions, often they are approximated using a constant factor for regulatory accounting purposes. The techniques discussed in this document could likely be revised to accommodate a variety of loss accounting practices; however, since the general apportionment solver is not intended to be a hydrologic simulator, specialized models may be required to implement more advanced management practices.

Requirements for Feasibility & Uniqueness

We want the solver to reliably provide a solution, which means we need to ensure that the problem is defined in a way that a solution exists. These are requirements for feasibility:

- Measured flows cannot be negative. (If the user supplies a negative flow measurement, the program will change it to zero.)
- Every flowline in the accounting flow network that connects a reach-node to a use-node or a reach-node to a storage-node must have an “unauthorized diversion” variable. Without these variables, the system of equations would be infeasible for each case a user diverts more water than is allowed by their rights.
- Every flowline in the accounting flow network that connects a storage-node to a reach-node or import water to a reach-node must also have a “spill” variable. Without these variables, the system of equations would be infeasible for each case where storage or import water is delivered to the stream but never diverted downstream.
- A flowline in the accounting flow network can have no more than one measurement.

We also want to define the problem in a way that ensures there is only one solution. These are requirements for uniqueness:

- All diversions need to be measured.
- All on-stream reservoirs need measurements for change in storage.
- All variables for water right diversions, changes, exchanges, storage deliveries, import deliveries, reservoir evaporation apportionments, and storage delivery loss apportionments need a relative priority value.

Program Input

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Program Output

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